

Top 100 Docker Scenario-Based Interview Questions

For Senior Backend / Product Company / FAANG-Level Preparation

Container Lifecycle

1. Your container exits immediately after startup. How do you troubleshoot?
2. Container is running but application is not accessible.
3. Container keeps restarting every few minutes.
4. Container works locally but fails in production.
5. Container randomly stops once every few days.
6. How do you determine whether the problem is Docker, OS, or application?
7. Container shows healthy but users receive errors.
8. Container receives SIGTERM during deployment. How should the application behave?
9. How do you achieve graceful shutdown?
10. How would you investigate a container that disappeared unexpectedly?

Dockerfile & Image Design

11. Your Docker image size is 3GB. How do you reduce it?
12. Why use multi-stage builds?
13. Build time increased from 2 minutes to 15 minutes.
14. How does Docker layer caching work?
15. Why should frequently changing files be copied later?
16. Difference between ADD and COPY in real projects?
17. Security team rejects your image because it runs as root.
18. Secrets accidentally became part of image layers.
19. How do you pass environment-specific configuration?
20. How would you version Docker images in production?

Resource Management

21. Container memory usage continuously grows.
22. How do you identify JVM memory leak inside Docker?
23. Container gets OOMKilled.
24. Difference between JVM OutOfMemoryError and OOMKilled.
25. CPU suddenly reaches 100%.
26. Application performance drops after moving to Docker.
27. Why should CPU and memory limits be configured?
28. Container is throttled despite low traffic.
29. How do Docker cgroups work?
30. What metrics would you monitor?

Storage & Volumes

31. Database container deleted accidentally.
32. How do you ensure database data survives?
33. Volume corruption is suspected.
34. Difference between Bind Mount and Named Volume.
35. Container logs fill the disk.
36. How would you backup Docker volumes?

- 37. How would you restore Docker volumes?
- 38. Application uploads files to container filesystem.
- 39. Container restart causes data loss.
- 40. Production server disk is full. Docker is suspected.

Networking

- 41. Container can access internet but cannot access another container.
- 42. Two services communicate via container IP and fail after restart.
- 43. Why use service names instead of IP addresses?
- 44. How does Docker DNS work?
- 45. Difference between bridge, host and overlay networks.
- 46. Database accidentally exposed to public internet.
- 47. Container receives traffic but application does not.
- 48. How would you debug networking inside a running container?
- 49. Inter-container communication intermittently fails.
- 50. How would you secure container-to-container communication?

Logging & Monitoring

- 51. Production issue occurs but logs are missing.
- 52. Why is docker logs insufficient at scale?
- 53. How would you collect logs from 1000 containers?
- 54. Application logs show nothing but users see failures.
- 55. What should be logged by the application versus Docker?
- 56. Container logs consume 500GB disk space.
- 57. How would you centralize logs?
- 58. How do you correlate logs across microservices?
- 59. What is structured logging?
- 60. How would you investigate intermittent failures?

Security

- 61. Container runs as root. Why is it risky?
- 62. How do you scan images for vulnerabilities?
- 63. How would you secure Docker daemon?
- 64. Developer stores AWS credentials inside image.
- 65. How should secrets be managed?
- 66. Why avoid using latest tag?
- 67. How do you verify image authenticity?
- 68. How would you harden a production Docker image?
- 69. Container escape vulnerability is discovered.
- 70. How do namespaces improve security?

Docker Compose

- 71. Spring Boot starts before PostgreSQL.
- 72. Compose services start successfully but application fails.
- 73. How do services discover each other?
- 74. One Compose service crashes repeatedly.
- 75. How would you scale services in Compose?
- 76. How do you manage environment variables across environments?
- 77. How would you deploy same Compose setup in Dev, QA and Prod?
- 78. How do restart policies work?

- 79. How do you debug Compose networking?
- 80. What are Compose limitations?

CI/CD & Production

- 81. Image passes testing but fails after deployment.
- 82. How do you rollback a bad deployment?
- 83. How do you achieve zero-downtime deployment?
- 84. How would you handle image promotion across environments?
- 85. Why should Dev and Prod use same image?
- 86. How would you implement Blue-Green deployment?
- 87. How would you implement Canary deployment?
- 88. Production deployment takes 20 minutes because images pull slowly.
- 89. Registry becomes unavailable.
- 90. How do you design a Docker deployment pipeline?

Advanced Production Scenarios

- 91. Traffic suddenly doubles. What Docker bottlenecks do you investigate?
- 92. One microservice starts causing cascading failures.
- 93. A containerized Java application becomes slow after 15 days uptime.
- 94. Server reboot occurs. How do services recover automatically?
- 95. Container image contains a critical CVE. What is your response plan?
- 96. How would you migrate a legacy VM application to Docker?
- 97. How would you run multiple versions of the same service simultaneously?
- 98. How would you troubleshoot a memory leak that only occurs in production?
- 99. How would you containerize a monolithic Java application?
- 100. Describe the complete debugging process when users report high latency, some containers restart, database connections fail, CPU usage is high, and logs show no obvious errors.